WSPRadar.org reconstruction & comparison

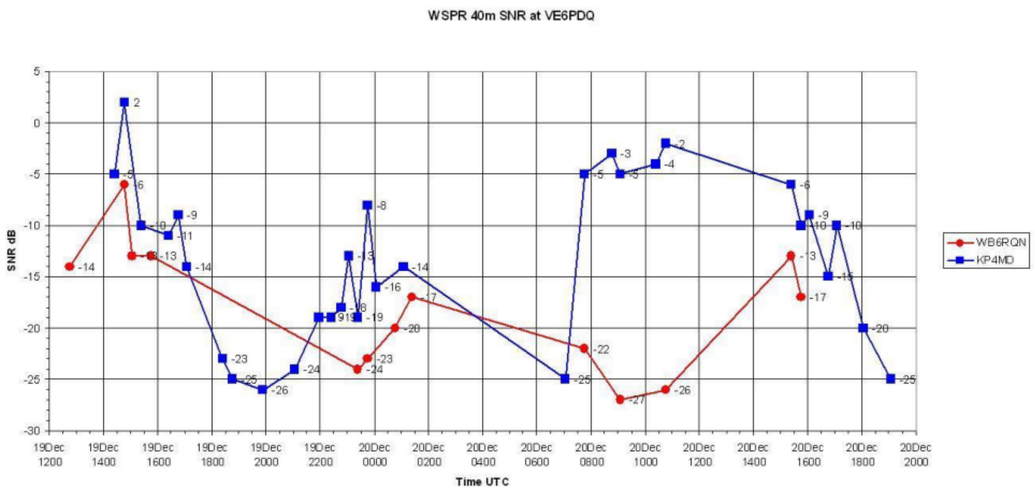
Reconciled RX direction: VE6PDQ transmitting to KP4MD / WB6RQN

Referenced publication: Carol F. Milazzo / KP4MD (2011). *Using the Weak Signal Propagation Reporter Network to Compare Antenna Performance.*

Source figure: **Figure 6** · Amateur-radio technical article and club presentation · 2011

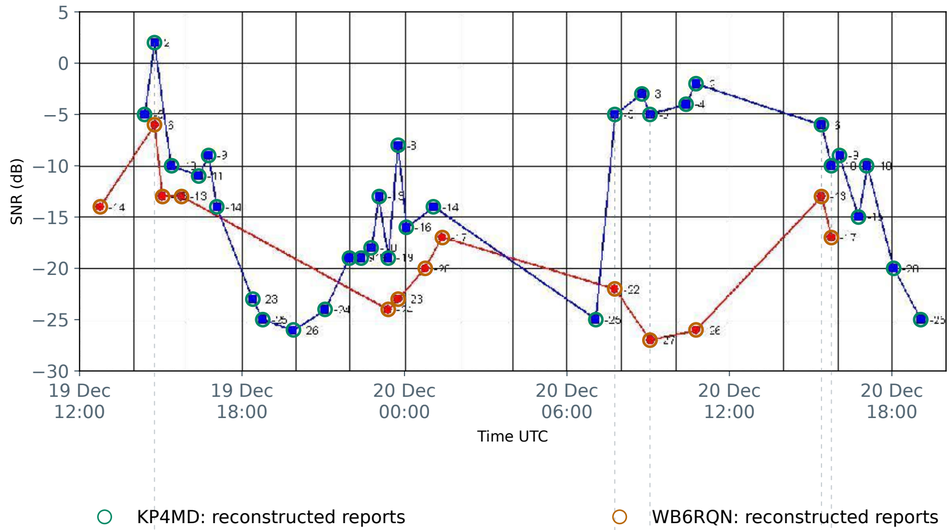
Demo: Historical RX replay | 40 m | 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC | VE6PDQ (DO34 / DO34IR) to KP4MD / WB6RQN | Reference correction 0 dB

Panel A - Original image from publication



Original image retained; matched reports establish the direction shown in the header.

Panel B - Reconstruction



44 paper anchors reconcile exactly in SNR. Panel B shows all 44 endpoint reports from the database in this path and time window.

All RX reports are 5 W. The common 37 dBm paper scale is the native 30 dBm value +7 dB.

Original dots and lines remain at full opacity beneath the colored circles. Exact database timestamps are used; no fitted time shift or SNR offset.

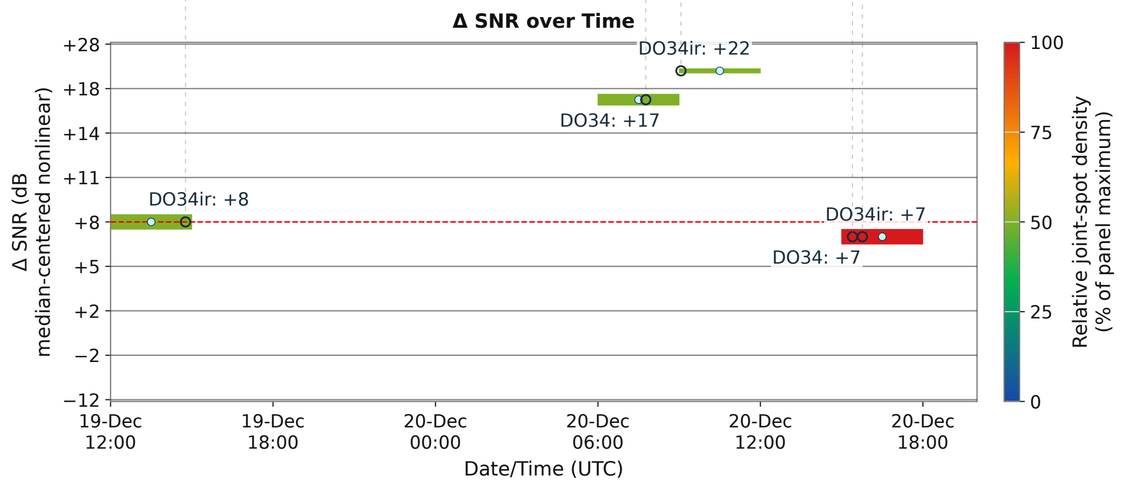
Panel C retains native 3h bins, density, median and IQR. Read the dB labels on its median-centered nonlinear axis; positive ΔSNR favors KP4MD.

Five pairs retain the separate DO34 and DO34IR identities.

Historical replay: observed coverage and differing operating schedules do not establish controlled antenna gain. Page 2 reconciles the native temporal coverage and counts.

Panel C - WSPRadar paired ΔSNR

5 Joint Spots | KP4MD - WB6RQN | exact full locator identity



Circles: exact Joint Spots; labels: full locator and ΔSNR. Dashed guides identify their two reports.

WSPRadar.org reconstruction & comparison

Observed RX coverage over time: counts and ΔDecode

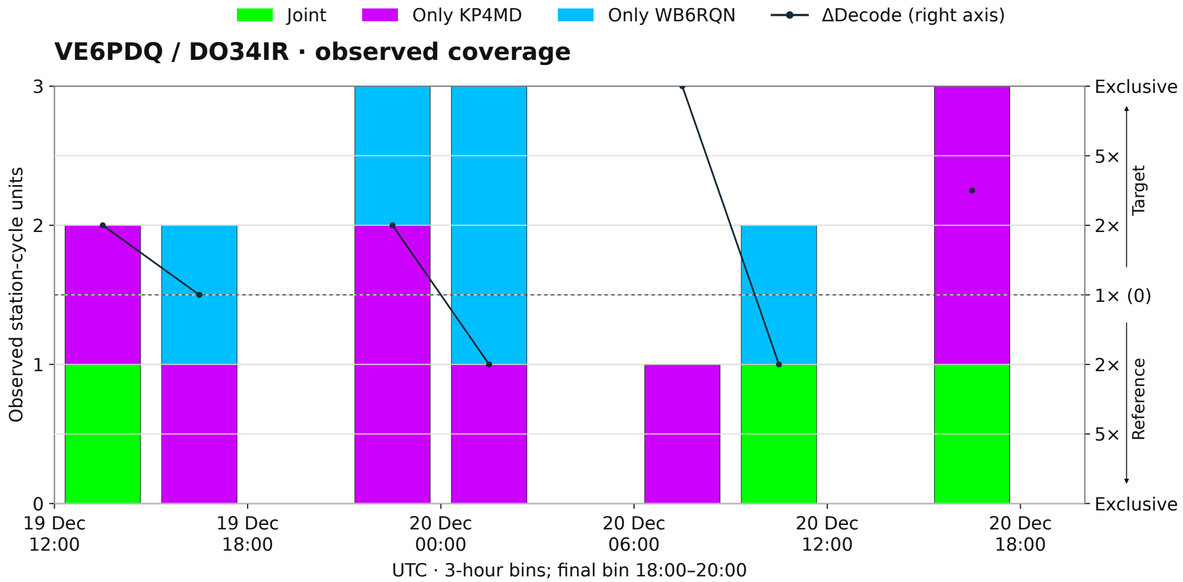
Referenced publication: Carol F. Milazzo / KP4MD (2011). *Using the Weak Signal Propagation Reporter Network to Compare Antenna Performance.*

Source figure: **Figure 6** · Amateur-radio technical article and club presentation · 2011

Demo: Historical RX replay | 40 m | 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC | VE6PDQ (DO34 / DO34IR) to KP4MD / WB6RQN | Reference correction 0 dB

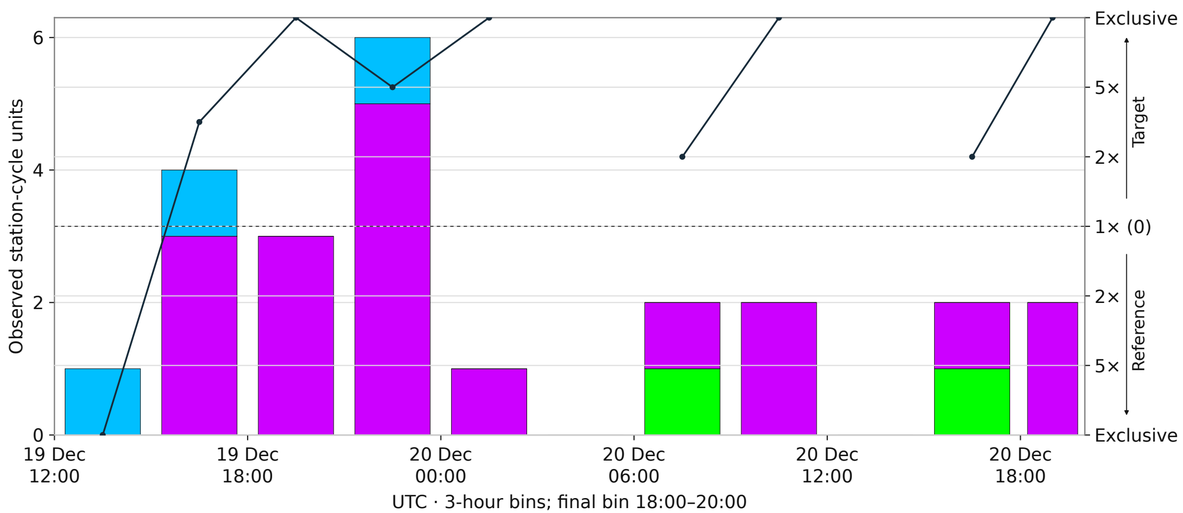
Overview · same-cycle, full-locator outcomes

Selected identity	Only T	Joint	Only R	T reports	R reports	Units	Reports	ΔDecode (×)
VE6PDQ / DO34IR	8	3	5	11	8	16	19	1.38×
VE6PDQ / DO34	18	2	3	20	5	23	25	4.00×
Selected path total	26	5	8	31	13	39	44	2.38×



VE6PDQ / DO34IR · exact bin counts

UTC bin	Only T	Joint	Only R	T	R	Units	ΔDecode (×)
19 Dec 12:00–15:00	1	1	0	2	1	2	2.00×
19 Dec 15:00–18:00	1	0	1	1	1	2	1.00×
19 Dec 18:00–21:00	0	0	0	0	0	0	
19 Dec 21:00–00:00	2	0	1	2	1	3	2.00×
20 Dec 00:00–03:00	1	0	2	1	2	3	0.50×
20 Dec 03:00–06:00	0	0	0	0	0	0	
20 Dec 06:00–09:00	1	0	0	1	0	1	∞×
20 Dec 09:00–12:00	0	1	1	1	2	2	0.50×
20 Dec 12:00–15:00	0	0	0	0	0	0	
20 Dec 15:00–18:00	2	1	0	3	1	3	3.00×
20 Dec 18:00–20:00	0	0	0	0	0	0	



VE6PDQ / DO34 · exact bin counts

UTC bin	Only T	Joint	Only R	T	R	Units	ΔDecode (×)
19 Dec 12:00–15:00	0	0	1	0	1	1	0.00×
19 Dec 15:00–18:00	3	0	1	3	1	4	3.00×
19 Dec 18:00–21:00	3	0	0	3	0	3	∞×
19 Dec 21:00–00:00	5	0	1	5	1	6	5.00×
20 Dec 00:00–03:00	1	0	0	1	0	1	∞×
20 Dec 03:00–06:00	0	0	0	0	0	0	
20 Dec 06:00–09:00	1	1	0	2	1	2	2.00×
20 Dec 09:00–12:00	2	0	0	2	0	2	∞×
20 Dec 12:00–15:00	0	0	0	0	0	0	
20 Dec 15:00–18:00	1	1	0	2	1	2	2.00×
20 Dec 18:00–20:00	2	0	0	2	0	2	∞×

T = Only T + Joint; R = Only R + Joint; table ΔDecode = T / R: 1.00× equal, ∞× Target only, 0.00× Reference only; blank = unobserved. Right-axis factors name the favored side.

Historical replay: only observed outcomes are counted; cycles with neither endpoint decoded are unobserved. Operating schedules and path conditions prevent interpreting coverage imbalance as antenna gain.

WSPRadar settings: peer distance < 5 000 km; minimum Joint Spots per station: 1; all directions; solar state: all; special-callsign and moving-station exclusions: off.